

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

--

CT
GROUP

1	6	S		
---	---	---	--	--

INDEX
NUMBER

--	--	--	--

BIOLOGY

9744/02

Paper 2 Structured Questions

15 September 2017
2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the question paper.

The use of an approved scientific calculator is expected, where appropriate.

All workings and appropriate units must be shown.

The number of marks is given in brackets [] at the start of each question or part question.

For Examiner's Use	
1	/ 8
2	/ 11
3	/ 18
4	/ 8
5	/ 12
6	/ 8
7	/ 12
8	/ 11
9	/ 12
Total	/ 100

Answer **all** questions in this paper.

Question 1 [8 marks]

Fig. 1.1 illustrates an electron micrograph of a plant cell.

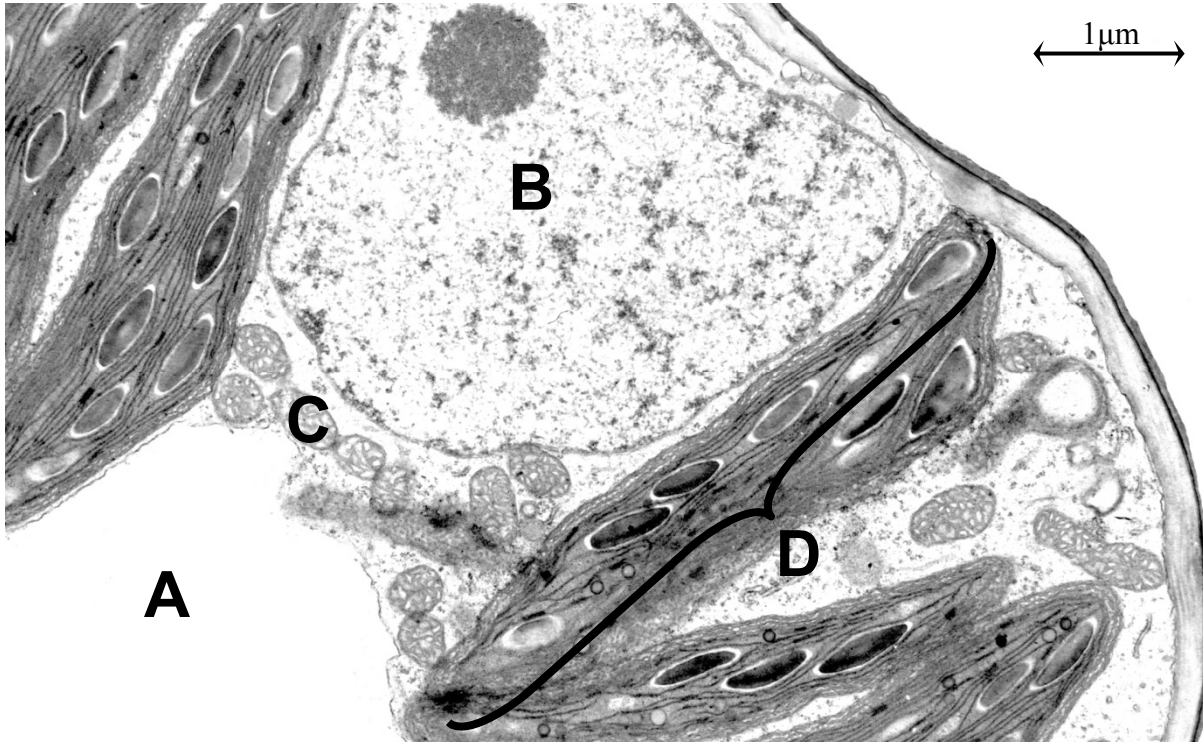


Fig. 1.1

- (a) With reference to **Fig. 1.1**, identify organelles labelled **A** and **B** and justify your answers. [4]

For A:

- Vacuole;;
- Large single membrane bound organelle at the centre of cell;;

For B:

- Nucleus;;
- Presence of a darkly stained spherical structure which is the nucleolus OR Presence of darkly stained regions in the nucleus which is heterochromatin;;

- (b) Explain how the structural features of organelles **C** and **D** can contribute to its role in the plant cell. [2]

- Presence of infoldings (cristae) in mitochondrion (organelle C) increases surface area for the attachment of enzymes, electron transport chains and ATP synthase;
- thus increasing its efficiency in oxidative phosphorylation (part of aerobic respiration);
- In chloroplast (organelle D), presence of thylakoids (which stack up to form grana with intergranal lamellae between the grana) increases surface area for packing of more photosystems and ETCs to maximize absorption of light;
- thus increasing the efficiency of light-dependent reactions of photosynthesis;

Fig. 1.2 is a diagram of the Golgi body, an organelle found in most eukaryotic cells.

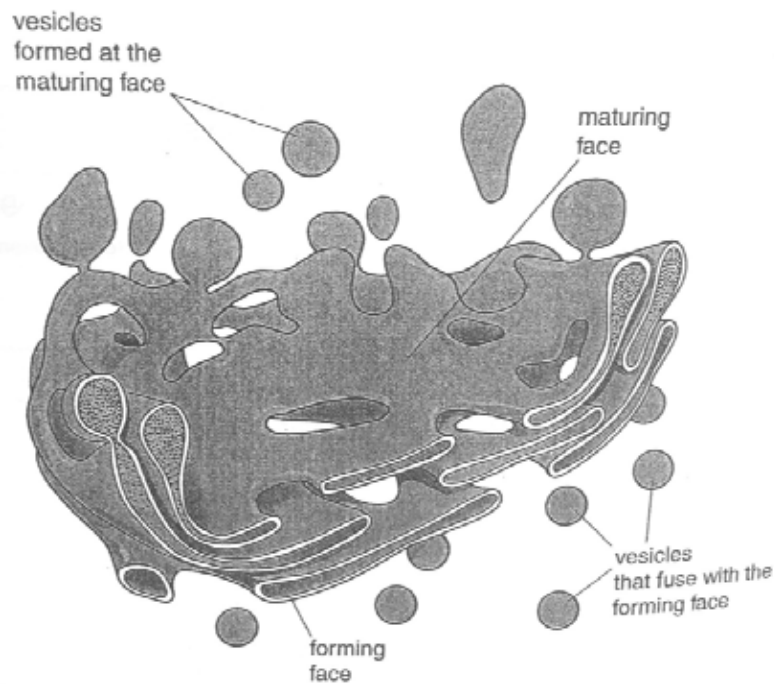


Fig. 1.2

(c) Besides serving as secretory vesicles, outline another role for the vesicles that are formed at the maturing face of the Golgi body. [2]

- a. (mention of) lysosomes;
- b. that contain hydrolytic enzymes / acid hydrolases;

choose any two:

- c. (ref. to intracellular digestion) – fusion with autophagosomes / food vesicles and subsequent hydrolysis / degradation into substances for cell use;
- d. (ref. to autophagy) aid in the removal of worn-out organelles;
- e. (ref. to autolysis) aid in cell killing / death, when hydrolytic enzymes are released into the cytoplasm;

Question 2 [11 marks]

Fig. 2.1 shows the structure of protease.



Fig. 2.1

(a) With reference to **Fig. 2.1**,

(i) explain why an enzyme acts only on a specific substrate. [2]

- a. An enzyme has an active site whose 3D conformation is complementary to the protein / polypeptide / peptide bonds it binds and acts on;
- b. The spatial arrangement of the contact and catalytic residues restrict the type of substrates it can catalyse;
- c. As only certain substrates have chemical groups orientated in a manner that would allow formation of temporary bonds with the contact residues to form ES complex;
- d. (Ref. to ES complex) Substrates must also have chemical groups that are orientated near catalytic residues to facilitate the breaking and reforming of bonds / conversion of substrate to product;

(ii) explain what determines the three-dimensional conformation of protease. [2]

- a. Unique sequence of amino acids determines bonds and interactions of R groups
- b. Segments of polypeptide chain is coiled into alpha-helices and folded into beta-pleated sheets
- c. Secondary structures fold back on themselves to form spherical / globular structure
- d. Tertiary structure stabilised by 4 types of bonds between R groups – hydrogen bonds, disulphide bridges, hydrophobic interactions and ionic bonds (quote any 2)

Sorghum is a staple food in Africa, but the major storage protein that it contains, kafirin, is not easily digested by protease enzymes. Upon heating, kafirin can undergo unfolding as shown in **Fig. 2.2**.

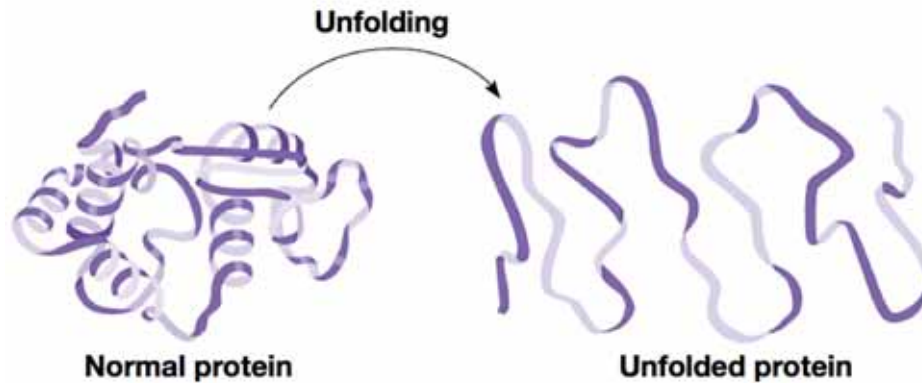


Fig. 2.2

The digestibility of the protein in two varieties of sorghum was measured when raw, after cooking with and without acid. Digestibility was measured as the percentage of the protein that would be broken down to amino acids during digestion.

The results are shown in **Fig. 2.3**.

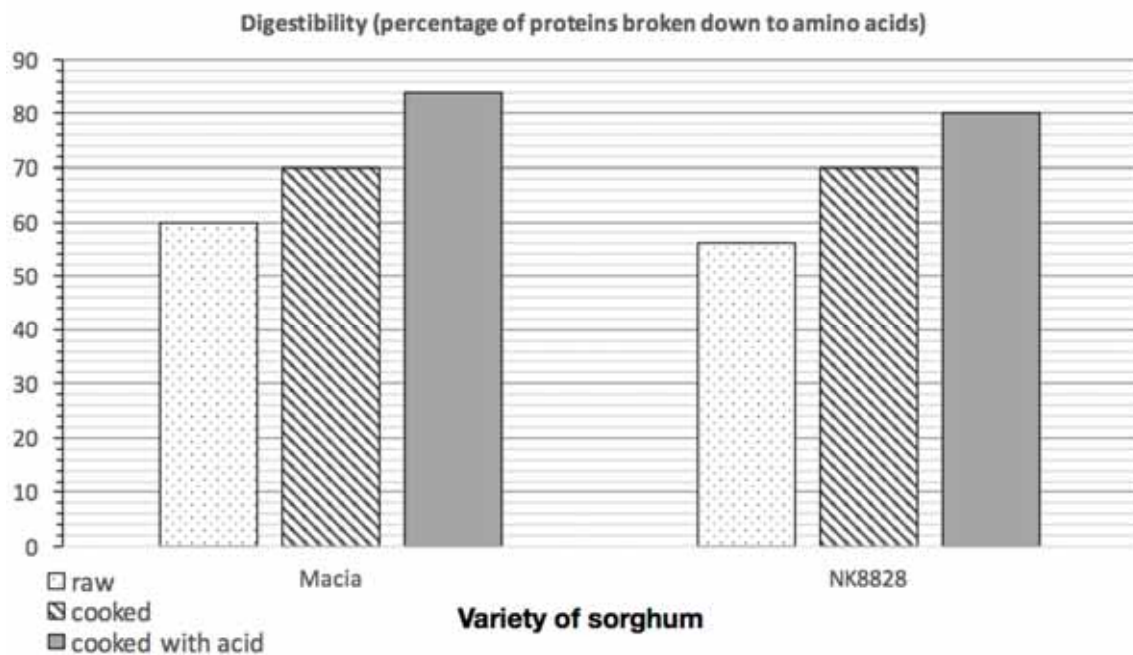


Fig. 2.3

(b) With reference to **Fig. 2.3**,

(i) **difference 1**: compare the digestibility of raw and cooked sorghum protein. [1]

a. **Cooked protein is more digestible than raw protein;**

b. **(Quote data) Macia: 60% digestibility in raw while 70% in cooked OR NK8828: 56% digestibility in raw while 70% in cooked;**

(ii) **difference 2**: compare the digestibility of cooked sorghum with and without acid. [1]

- a. Cooked protein with acid is more digestible;
- b. (Quote data) Marcia: 70% digestibility in cooked while 84% when cooked with acid
OR NK8828: 70% digestibility in cooked while 80% when cooked with acid;

(iii) and using **Fig. 2.2** with your knowledge of protein structure and enzyme activity, account for the two differences described in parts (i) and (ii). [3]

Difference between raw and cooking

- a. Cooking involves heating at high temperatures
- b. -> excessive molecular motion may break the bonds present in kafirin

Differences between cooking with and without acid

- c. Presence of high concentration of H^+ in acid
- d. -> COO^- groups of amino acid residues in Sorghum may be neutralised to $COOH$
OR
- e. -> $C=O$ groups of the peptide bond may also be protonated / or $C=O$ becomes positively charged

Both results in..

- f. e.g. disruption / breakage of hydrogen bonds and ionic bonds -> denaturation
- g. This results in a change in secondary and tertiary structure, and the protein may become less compacted
- h. Protease can now easily access and bind to the polypeptides -> increased digestibility

(c) Describe how the bond holding the two amino acids together may be broken to release the two amino acids. [2]

- a. hydrolysis of peptide bond
- b. water molecule is added
- c. hydroxyl group will allow the reformation of carboxylic group ($-COOH$) in one amino acid
- d. the other hydrogen will be gained by the other amino acid to reform its ($-NH_2$) amino group.
- e. Hydrolytic enzymes may also be involved

Max 2m

Question 3 [18 marks]

The production of membrane-bound or secreted immunoglobulin M (IgM) depends on whether the B cell is activated. Naïve B cells produce membrane-bound IgM while activated B cells produce soluble IgM that is secreted out of the cell.

In naïve B cells, the gene coding for heavy chain of IgM is expressed to produce a long RNA transcript, while the same gene codes for a shorter RNA transcript when the cell is activated (see **Fig. 3.1**). The two different types of RNA are processed differently, resulting in two different types of antibodies upon translation.

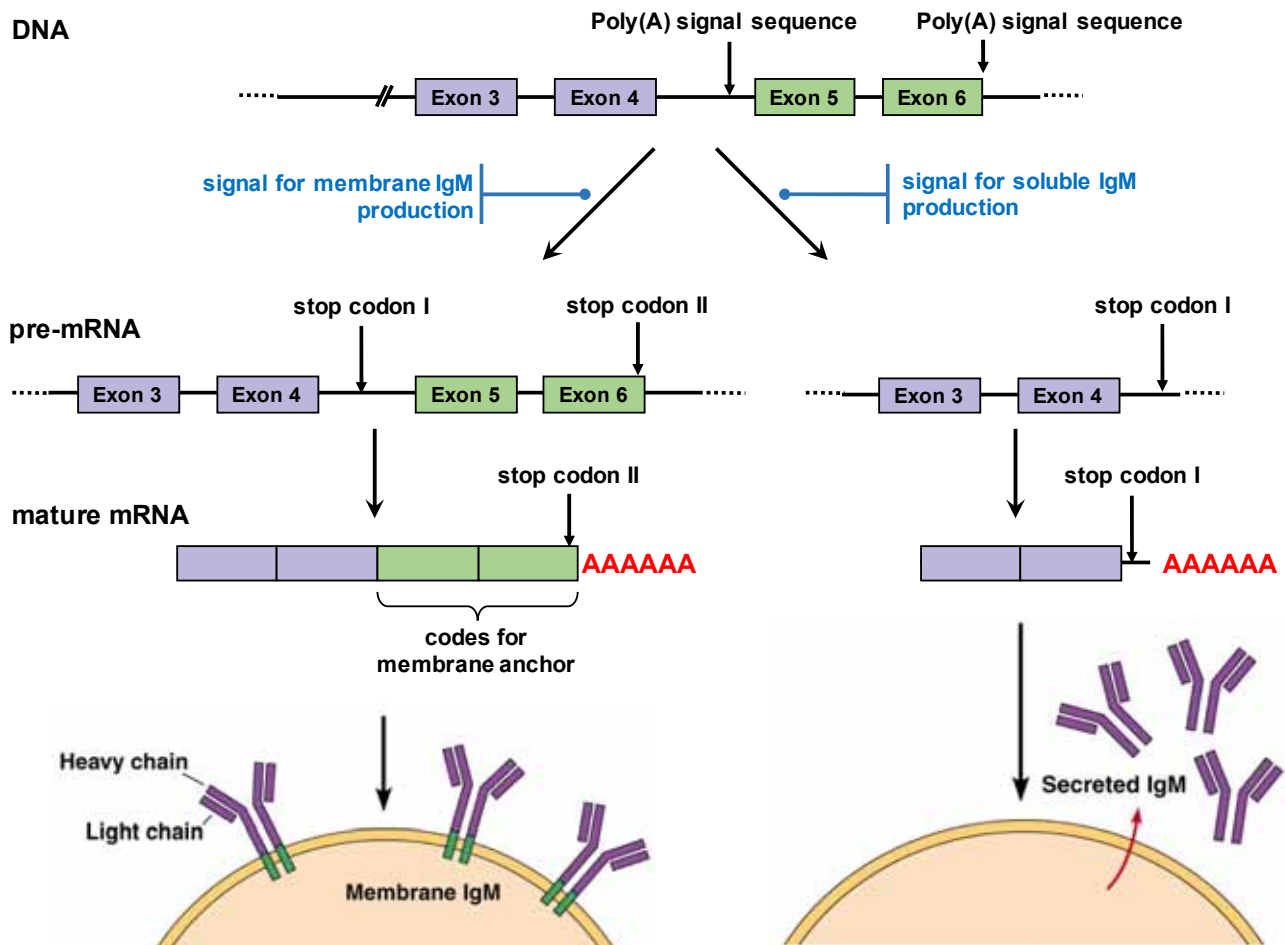


Fig. 3.1

Fig. 3.2 shows the region of DNA between exons 4 and 5.

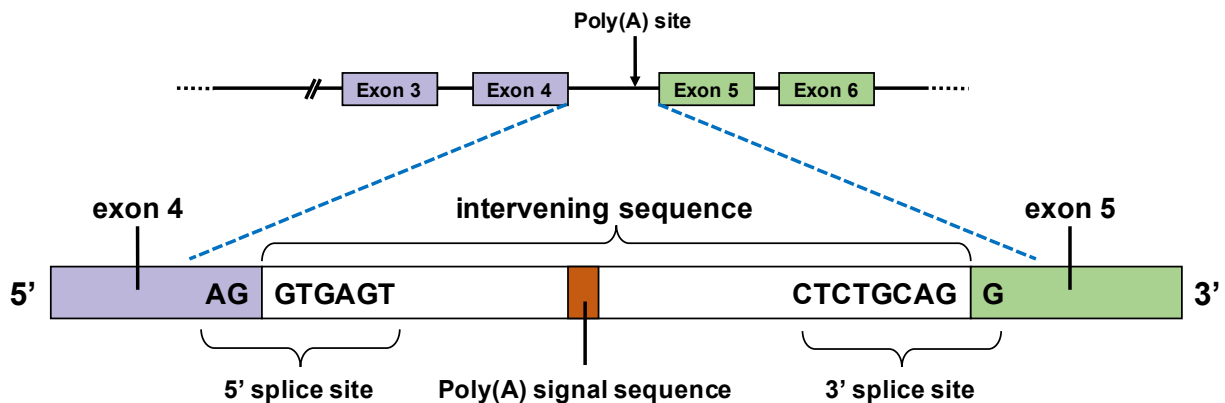


Fig. 3.2

(a) With reference to Fig. 3.1,

- (i) describe how the different types of mRNA produced determine how IgM is membrane-bound or secreted out of the cell. [2]
 - a. mature mRNA from naïve B cells contains exons 5 and 6/ORA;;
 - b. which upon translation results in an amino acid sequence coding for a membrane anchor (at the C-terminus) → allows anchorage of the IgM into the membrane;
 - c. Secreted antibodies lack this (membrane anchor) sequence → no anchorage onto PM → secreted;
- (ii) describe how IgM is held in the membrane of B cells. [2]
 - a. The surface of the protein next to the phospholipid fatty acid tails is made up of amino acids that are hydrophobic / with hydrophobic R groups;
 - b. Thus held in place via hydrophobic interactions with the hydrophobic fatty acid tails;
 - c. Amino acids forming the surface region of the protein adjacent to the phosphate heads are hydrophilic;
 - d. And thus held in place via hydrophilic interactions / hydrogen bonding;
- (iii) The membrane anchor in membrane IgM is reported to be associated with several proteins on the cytoplasmic side.

Suggest the function of these associated proteins. [1]

Choose any answer:

- a. **relay proteins that transduce signals within the cell upon activation of IgM / binding of antigen to IgM / resulting in receptor-mediated endocytosis ;;**
- b. **proteins that stabilise the IgM antibody onto the PM ;;**

- (b) The activation of a naïve B cell involved the recognition of specific antigens, and mediated by helper T cells.

With reference to **Fig. 3.1**,

- (i) describe the events occurring at the promoter of the IgM heavy chain gene locus in both naïve and activated B cells. [2]

- a. TBP recognises and binds to TATA box sequence (5'-TATAAA-3') at the promoter;
- b. Which then recruits the binding of other GTFs → subsequent recruitment of RNAPII to the promoter;
- c. Resulting in the formation of transcription initiation complex (TIC);
- d. Local unwinding and unzipping of DNA at transcriptional start site;
- e. Transcription of antibody gene begins

No marks awarded for (e), but added in for coherence

- (ii) describe how transcription is terminated when the B cell is activated. [2]

Compulsory points:

- a. B cell activation → signal for soluble IgM present;
- b. RNAPII transcribes the polyadenylation sequence located downstream of exon 4;

Any two points:

- c. which is 5'-AAUAAA-3' on the RNA/ (DNA sequence);
- d. which serves as a recognition and binding site for endonucleases;
- e. that bind 10-35 nucleotides downstream of the polyadenylation sequence → cleaves RNA transcript from RNAPII → termination of transcription;

- (iii) explain how the mature mRNA produced is relatively more stable than the pre-mRNA. [3]

- a. due to 5' capping / addition of modified guanine at the 5' end of mRNA;
- b. catalysed by capping enzyme complex;
- c. as well as the addition of many adenine ribonucleotides at the 3' end of the mRNA;
- d. catalysed by poly(A) polymerases;
- e. both of which prevents / delays the degradation of mRNA by exonucleases;;

- (c) With reference to both **Fig. 3.1** and **3.2**, explain the importance of the 5' and 3' splice site. [2]

- a. sequences that flank the ends of the introns;
- b. that allow the recognition and binding of snRNPs to the splice sites;
- c. due to the complementary base pairing/H bond formation between the snRNA and splice site sequence;
- d. this further recruits other (accessory/mediator) proteins forming the spliceosome;
- e. allowing for the process of RNA splicing excising introns and splicing flanking exons;

Max 2

In order to understand the process of translation, researchers isolated a clone of activated B cells and introduced small interfering RNA (siRNA) into these cells. siRNA is an RNA molecule that can be designed and produced *in vitro*.

In this study, the siRNA introduced was complementary to a short segment of RNA on the 5' end of the mature mRNA. It was found that these activated B cells were unable to produce soluble IgM antibodies.

(d) (i) Describe the structural features of the monomers in siRNA. [2]

- a. monomers are ribonucleotides;
- b. composed of a 5C ribose sugar;
- c. linked to one nitrogenous base (A, U, G, C) at the C1' of the ribose sugar;
- d. and three phosphate groups at the C5' of the ribose sugar;

(ii) Suggest why the introduction of the siRNA did not result in production of IgM antibodies. [2]

- a. prevents the process of translation;;
- b. by preventing the recognition of the small ribosomal subunit at the 5' end of mRNA;
- c. and thus the formation of the translation initiation complex / recruitment of the large ribosomal subunit;

Question 4 [8 marks]

Colon cancer begins when a number of epithelial cells lining the colon proliferate excessively due to changes in the several genes. **Fig. 4.1** shows the progression of colon cancer from a normal epithelium to a metastatic colon cancer.

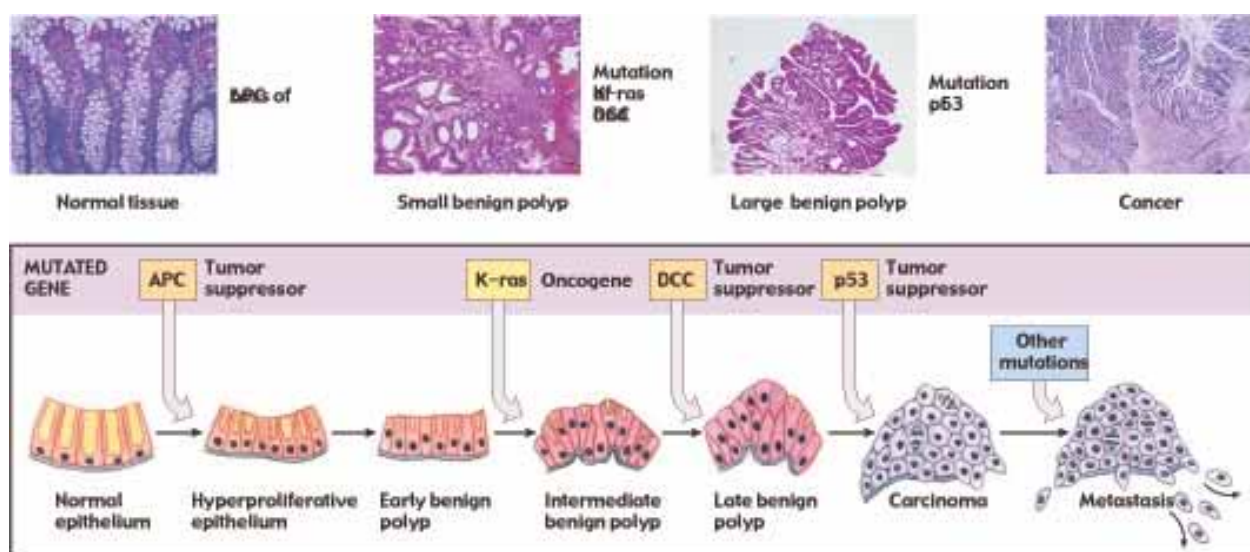


Fig. 4.1

- (a) With reference to **Fig. 4.1**, explain how this supports the multi-step model of cancer. [2]
- a. a single mutation is not sufficient to transform a normal cell to a cancerous one / to cause cancer / multiple mutations required to transform a normal cell to a cancerous one;
 - b. (quote) 4 different mutations in tumour suppressor genes (DCC, p53, APC) and proto-oncogenes (K-ras);
 - c. (mention of) LOF of TSG and GOF of proto-oncogene into oncogene;
 - d. each of these mutations contribute to cancer formation by providing the cell with greater proliferative capabilities / AW;
 - e. other mutations also result in the tumour gaining invasive / metastatic capabilities → spread of cancer;

(Quotation can appear anywhere in the answer)

Max 2m

The effectiveness of anti-cancer drugs may be determined by growing different tumours in culture.

The effectiveness of two drugs on two human tumours (**A** and **B**) from different tissues was assessed. The two drugs, T138067 and vinblastine, were added to the tumours in the culture on days 5, 12, and 19. The volumes of the tumours were compared with the volumes of tumours that were not treated with any drugs.

The results are shown in **Fig. 4.2**.

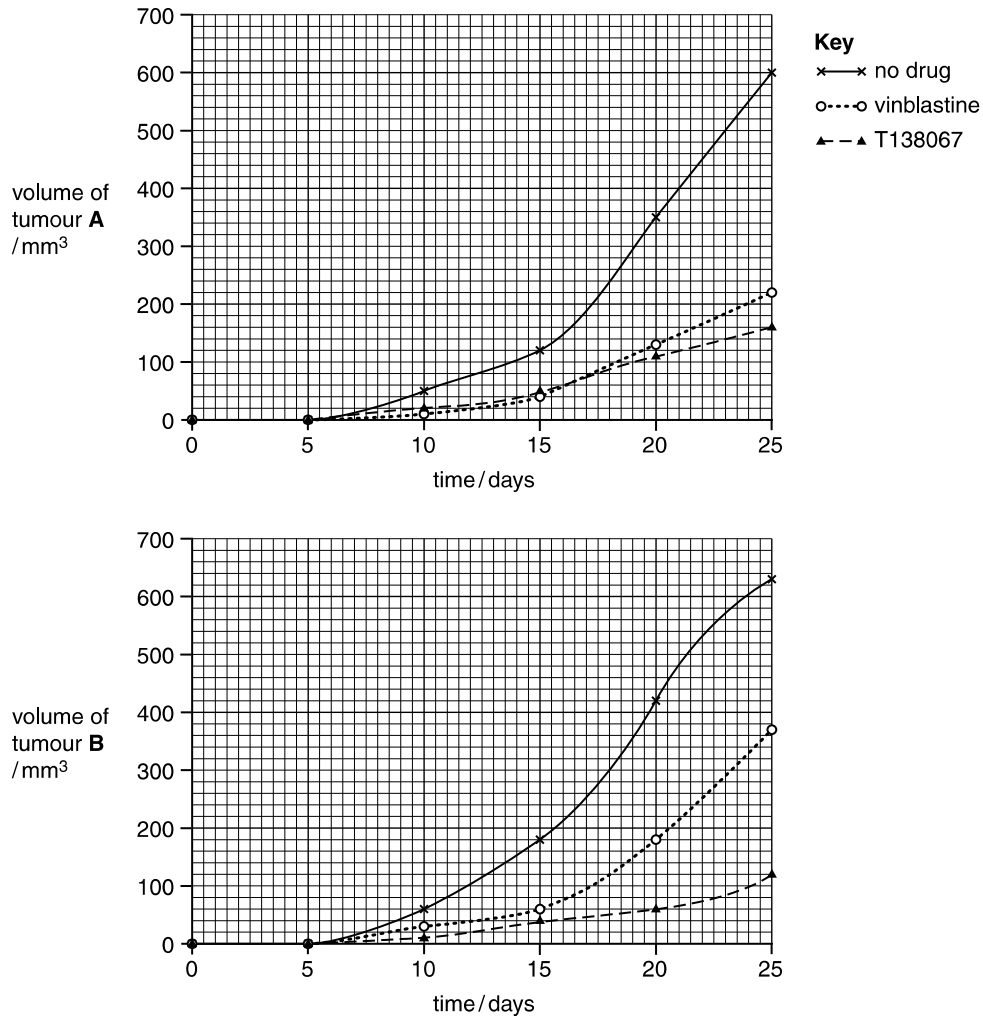


Fig. 4.2

- (b) Using the data in **Fig. 4.2**, compare the effectiveness of the two drugs used to treat the tumours. [3]

Choose any 3 pairs of answers:

- a. both drugs are effective in treating tumours (compared to no drug);
- b. (provide comparative data quote, both drugs compared to no drug);
- c. T138067 more effective than vinblastine against both tumours (A and B);
- d. relevant comparative data quote; e.g. volume of 220 v 160 mm^3 at day 25 for tumour A
- e. little difference in effectiveness between vinblastine and T138067 against tumour A up to day 18 / AW;
- f. ref. similar effectiveness against tumour B until after day 15 ;
- g. ref. to effectiveness of both drugs detectable from about 7–10 days / AW;
- h. both drugs, not completely effective in stopping growth / tumours continue to grow;
- i. AVP ;; e.g. greater effectiveness of, T138067 with B / vinblastine with A with quotation

- (c) Vinblastine disrupts the formation of the spindle apparatus during mitosis.

Explain how vinblastine has its effect as an anti-cancer drug. [3]

- a. growth of tumour involves mitosis / mitotic cell division;
- b. which involves the assembly and disassembly of microtubules / spindle apparatus / AW;

choose any two (from c to e):

- c. vinblastine prevents the attachment of spindle fibres to the chromosomes at the kinetochore / centromeres in prophase;
- d. as well as in aligning the chromosomes at metaphase;
- e. and in the separation of sister chromatids at anaphase;
- f. mitosis cannot proceed → cell cycle is arrested;
- g. thus preventing growth of tumours;
- h. AVP / arrested cells then undergo apoptosis → reduce tumour volume;

Question 5 [12 marks]

Fig. 5.1 shows an electron micrograph of H1N1.

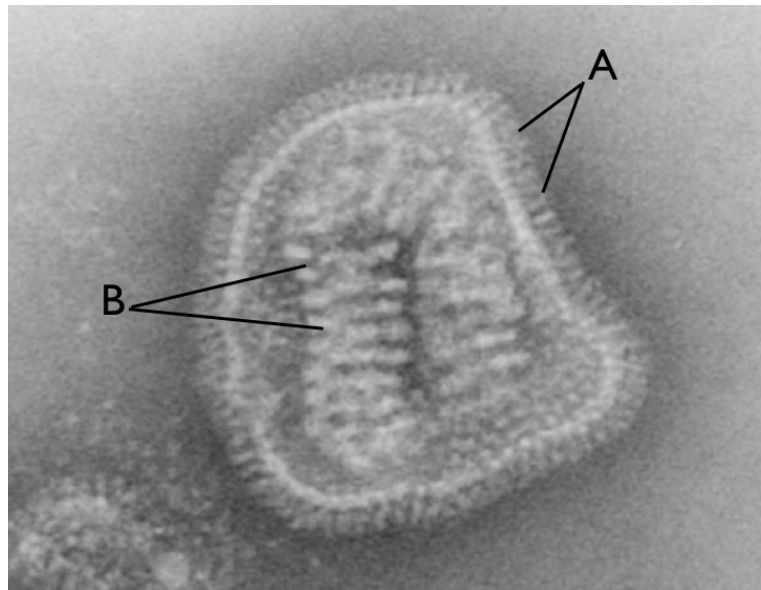


Fig. 5.1

(a) Identify the components labelled **A** and **B**. [3]

A

- a. Neuraminidase;;
- b. Haemagglutinin;;

B

- c. RNA segments; negative / antisense;

(b) RNA-dependent RNA polymerase can be found in this virion while reverse transcriptase can be found in retrovirus.

Besides the type of virus, describe one similarity and two differences between RNA-dependent RNA polymerase and reverse transcriptase. [3]

Similarity:

- a. Both are viral proteins, i.e. they are only coded for by viral genes and not the host genes.
- b. Both are dependent on RNA as a template

Differences

Points	RdRp	Reverse Transcriptase
c. Template used	RNA (both positive and negative strand)	RNA template, then using the single-stranded cDNA
d. Products	Only formation of RNA – sense RNA for translation when using negative RNA as a template , and negative RNA when using positive	Formation of cDNA-RNA strand using RNA as template, formation of double stranded cDNA when using cDNA as a template

	RNA as a template to create more copies of viral genome	
	Single stranded RNA	Double stranded cDNA
e. Catalytic activity	RNA polymerase – extension of RNA using host free ribonucleotides	Reverse transcription – extension of cDNA Ribonuclease/RNase – degrade RNA in cDNA-RNA DNA polymerase – formation of complementary DNA to form double-stranded cDNA
f. Monomers	Ribonucleotides	Deoxyribonucleotides

Oseltamivir, better known under its trademark name as Tamiflu, is a recommended medication for H1N1. Its mechanism of action is illustrated in **Fig. 5.2** and **Fig. 5.3**.

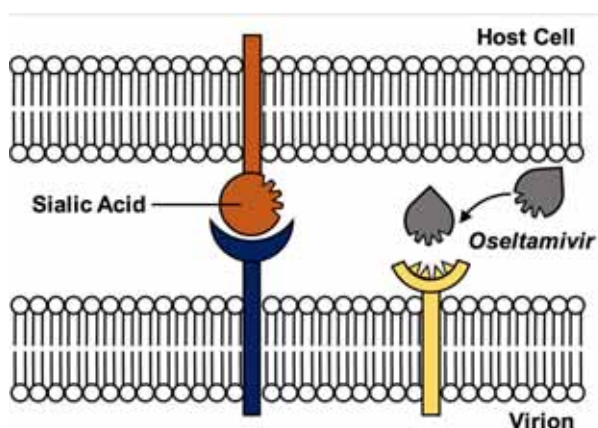


Fig. 5.2

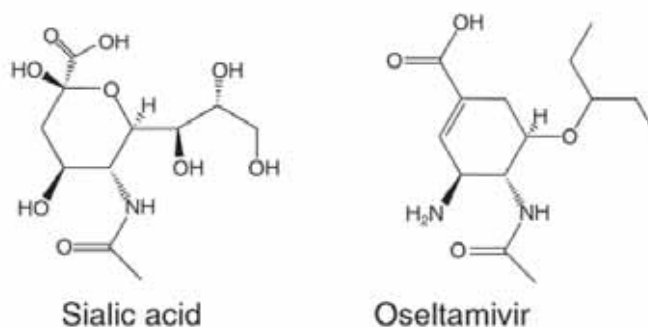


Fig. 5.3

(c) Explain how oseltamivir affects the reproduction cycle of H1N1. [2]

- Oseltamivir has similar conformation as sialic acid
- And thus acting as an inhibitor
- It is thus able to bind to the active site of neuraminidase
- This prevents neuraminidase from cleaving sialic acid
- Hence virion will not be able to be released via budding to infect other cells, (preventing the virus from spreading to the rest of the respiratory tract)

Many studies were conducted to determine the effectiveness of Oseltamivir.

In one of these studies, the effects of oseltamivir were studied on patients infected with influenza virus. These patients were randomly separated into two groups – one group administered with a placebo (a pseudo-drug with no effect) while the other group was administered oseltamivir. Many symptoms were evaluated, with cough being one of the symptoms represented in **Table 5.4**.

Table 5.4

Symptoms	Study Groups	
	Placebo (n=129)	Oseltamivir, 75mg (n=124)
Cough		
Duration, hr	55	31
Severity, arbitrary units	110	67

Facing the wave of flu cases, oseltamivir was stocked up in many medical facilities across countries where there was widespread administration of this drug. Thereafter, there were reports of side effects that included hallucination, and a flurry of questions rose to challenge the validity of studies such as the one above.

- (d) Suggest what could have been done in the above study to increase the confidence of the effects of oseltamivir. [1]

- a. Increase sample size
- b. Match the participants
- c. Repeat the study with a different sample group

Any 1m for each point

- (e) In a separate case, a patient was treated for both H1N1 and H3N3. During this treatment, he was infected with H3N3. Upon testing, it was found that his blood now contains H3N3 and a new strain of virus, H1N3.

Explain how this new strain could have arisen. [3]

- a. The new strain of virus could have arisen due to antigenic shift.
- b. As H1N1 and H3N3 infect the same cell simultaneously, their protein capsids and lipid envelopes are removed, exposing their RNA.
- c. Influenza virus can undergo genetic shift because it contains a segmented genome composed of 8 different segments.
- d. The new strain, H1N3, could have arisen via genetic recombination of the genome segments.
- e. The random assembly of RNA segments from two different viruses
- f. resulting in a mixture of surface antigens of both H1N1 and H3N3 strains.

Question 6 [8 marks]

Fig. 6.1 shows two bacterial cells, bacterium **A** and bacterium **B** involved in process **X**.

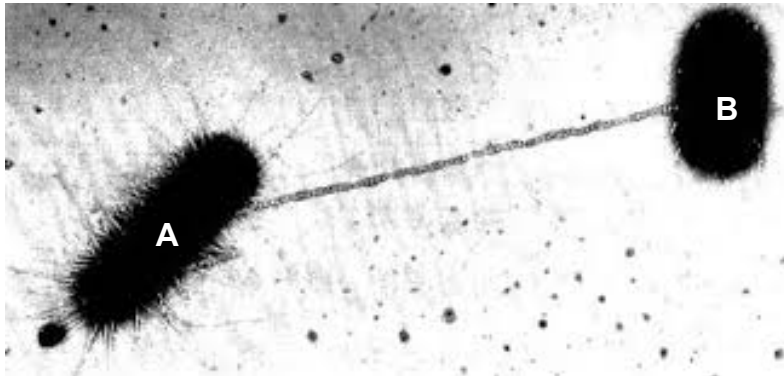


Fig. 6.1

(a) Describe the process **X** and elaborate on its significance. [3]

- a. Conjugation;;
- b. F+ donor cell synthesises sex pilus and make direct contact with F- recipient cell;
- c. Transfer of F plasmid from donor to recipient cell via formation of conjugation tube/cytoplasmic bridge (Not through sex pilus);
- d. Allows for genetic recombination to occur in bacteria/contributes to increased genetic variation in bacteria;
- e. Thus allows for bacteria to adapt to changing conditions;

Fig. 6.2 shows a section of an electron micrograph of bacterium **A** in **Fig. 6.1**.



C

Fig. 6.2

(b) Explain the role of structure C. [3]

- The F plasmid of the donor cell contains several genes that promote its transfer to other cells (*tra* genes)
- These genes code for the protein subunits that assemble on the surface of the bacterial cell which forms the sex pilus.
- The sex pilus extends and attaches to a recipient cell like a grappling hook.
- The sex pilus retracts and the 2 cells are pulled towards each other so that a temporary mating/conjugation bridge can be formed. (sex pilus $\neq$ conjugation bridge)
- The F plasmid also contain an origin of replication where DNA replication can initiate.
- The F plasmid also contains *oriT* whereby one strand of DNA can be transferred to the F- cell via rolling circle mechanism.
- The F plasmid may also contain beneficial genes such as antibiotic/xenobiotic resistance.

(c) State two ways in which structure C differs from the bacterial chromosome. [2]

Bacterial chromosome	F-Plasmid
a) Longer with more genes	a) Smaller with fewer genes
b) Contain essential genes which are necessary for bacterial's survival under normal conditions	b) contains genes that are not essential but are useful under specific / stressful environmental conditions
e.g. genes are responsible for production of enzymes for cell metabolism	e.g. genes code for antibiotic resistance ability , <i>tra</i> genes (DNA that encode the synthesis of sex pili)

Question 7 [12 marks]

An experiment was conducted to investigate how various factors affect the rate of photosynthesis in cabbage. **Fig. 7.1** below shows the results of the experiments conducted.

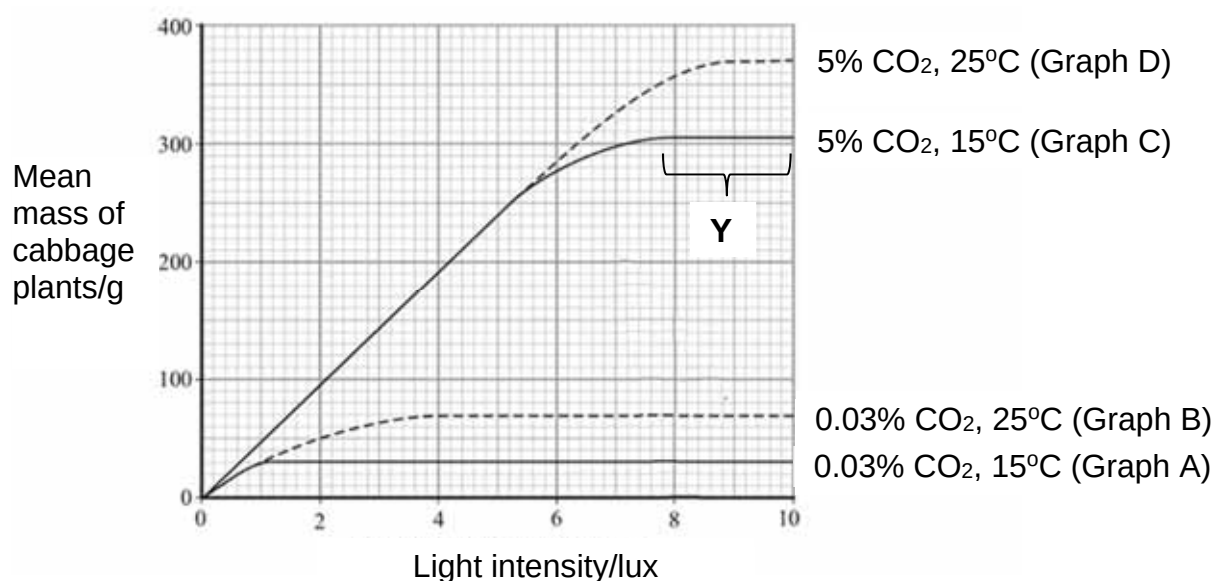


Fig. 7.1

(a) With reference to **Fig. 7.1**,

(i) state the best conditions for the growth of cabbage. [1]

5% carbon dioxide, 25°C, and (any value between 8.8-10) lux;;

(ii) explain the region marked Y. [2]

- a. **Light intensity is no longer the limiting factor / light saturation is achieved, ;**
- b. **as mean mass of cabbage plants remains constant at 305g (accept range between 300-310g) beyond 8 lux (accept range between 7.6-8) lux;**
- c. **Mean mass of cabbage plants will only increase if temperature is increased from 15°C to 25°C. ;**
- d. **Rate of photosynthesis is limited by temperature;**

(iii) describe and explain the effect of increasing carbon dioxide concentration on the mean mass of cabbage at 25 °C. [3]

- a. **As the carbon dioxide concentration increases from 0.03% to 5%, the maximum mass increases from 70g to 370g of cabbage;;**

Max 1.5 for marking points b) to e)

- b. **More CO₂ is used for more carbon fixation during light independent reaction;**
- c. **Increase in carbon dioxide concentration will increase the frequency of effective collisions between enzyme, Rubisco and substrates, RuBP and CO₂;**
- d. **Hence rate of enzyme-substrate complex formation increases;**
- e. **Rate of formation of glyceraldehyde-3-phosphate increases;**

R: product, need to mention which product is responsible for the increase in mass

- f. **More glyceraldehyde-3-phosphate molecules are converted to form more glucose and cellulose which increases the mass of lettuce;**

Compare Graph B and D only. Idea of increase/more and rate/per unit time should be included throughout

R: description of light-dependent reactions which does not explain why mean mass of cabbage increases

(iv) The average carbon dioxide content of the natural environment is 0.035%. Using this fact, and the information given in **Fig. 7.1**, what conclusion can be made about how carbon dioxide affects rate of photosynthesis in the natural environment? [2]

- a. **.035% in the natural environment is close to 0.03% in the experiment,**
- b. **shows that carbon dioxide concentration is a limiting factor on the rate of photosynthesis in the natural environment;;**
- c. **At maximum light intensity at 25°C, Graph D showed higher rate of photosynthesis which resulted in 370g mass of cabbage whereas**
- d. **Graph B showed lower rate of photosynthesis which resulted in only 70g of cabbage;;**
OR
- e. **At maximum light intensity at 15°C, Graph A showed higher rate of photosynthesis which resulted in 30g mass of cabbage whereas**

f. Graph C showed lower rate of photosynthesis which resulted in only 305g (accept range between 300-310) of cabbage;;

- (b) While photosynthesis is the process by which carbon dioxide and water are used as starting materials for the synthesis of glucose using light energy, respiration involves releasing chemical energy in organic molecules such as glucose by oxidation and made available to living cells in the form of ATP. In particular, the yield of ATP under aerobic and anaerobic respiration are very different.

Explain the small yield of ATP from anaerobic respiration in both yeast and animals. [4]

- During anaerobic respiration, there is no oxygen, thus there is no final electron acceptor for oxidative phosphorylation to take place;
- This will result in electron flow via electron transport chain being blocked;
- When this happens, there will be no protons pumped into intermembrane space from the matrix of mitochondria;
- Proton gradient (and hence proton motive force) will be dissipated;
- No chemiosmosis via ATP synthase → no ATP production via oxidative phosphorylation;
- NAD⁺ and FAD will not be regenerated so link reaction and Krebs cycle cannot take place to generate more ATP, NADH and FADH₂;
- Only glycolysis can take place to generate a net gain of 2 ATP molecules per glucose molecule;
- Anaerobic respiration via lactate fermentation or alcoholic fermentation does not produce ATP but only regenerates NAD⁺ for glycolysis to continue;
- only a net of 2 ATP molecules are synthesized via glycolysis via substrate level phosphorylation;
- a large amount of energy is locked in lactate / alcohol;

Question 8 [11 marks]

In cattle, coats may be solid white, solid brown, or beige. When true breeding solid whites are mated with true-breeding solid brown, the F1 generation consists of all solid white individuals. Mating among the F1 generation have resulted in the following ratio:

23 solid white
6 beige
3 solid brown

- (a) (i) Draw a genetic diagram of the above described cross. [4]

Let W represent white colour fur
Let w represent coloured fur
Let B represent beige fur
Let bb represent brown fur

F1 Phenotype	white		white	
F1 genotype	WwBb		WwBb	
Meiosis				
Gametes	WB	wB	wB	wb
Offspring genotype and phenotype	WB	WB	Wb	wB
	WB	WWBB	WWBb	WwBB

		white	white	white	White
	Wb	WWBb white	WWbb white	WwBb white	Wwbb white
	wB	WwBB white	WwBb white	wwBB beige	wwBb beige
	wb	WwBb white	Wwbb white	wwBb beige	wwbb brown
Offspring phenotypic ratio	12 white : 3 beige : 1 brown				

- Correct legend;
- Correct F1 genotype
- correct genotype;
- correct gametes;
- gamete circle and arrows;
- punnett square correct genotypes;;
- punnett square correct phenotypes;;
- phenotypic ratio;

(ii) Explain what is meant by 'epistasis' in this context. [3]

- Epistasis = two protein products interacting in the same molecular pathway to affect a characteristic;;
- Dominant epistasis
- W is epistatic to B and/or b alleles.
- dominant allele W could code for an enzyme that catalyses the breakdown/destroys the brown or beige pigment OR that destroys/denatures an enzyme that is necessary for pigment formation OR inhibitor that blocks synthesis of brown or beige pigments; OR inhibitor that prevents deposition of any colour pigment to be deposited on the fur
- As long as you have a dominant W allele, it will mask the expression of B/b gene locus (this is an eg of dominant epistasis)

(iii) The chi-squared test was later performed on the F2 data.

Explain what a chi-squared test is used for. [1]

- It is a valuable tool to estimate the probability if whether the deviation between observed and expected results is significant or not
- It allows one to determine if the deviation between observed and expected results is due to random change or wrong prediction

Marfan syndrome is a genetic disorder of the connective tissues. People with Marfan tend to be tall, and thin, with long arms, legs, fingers and toes. The most serious complications involve the heart and aorta.

Fig. 8.1 shows the inheritance of Marfan syndrome over three generations in an extended family.

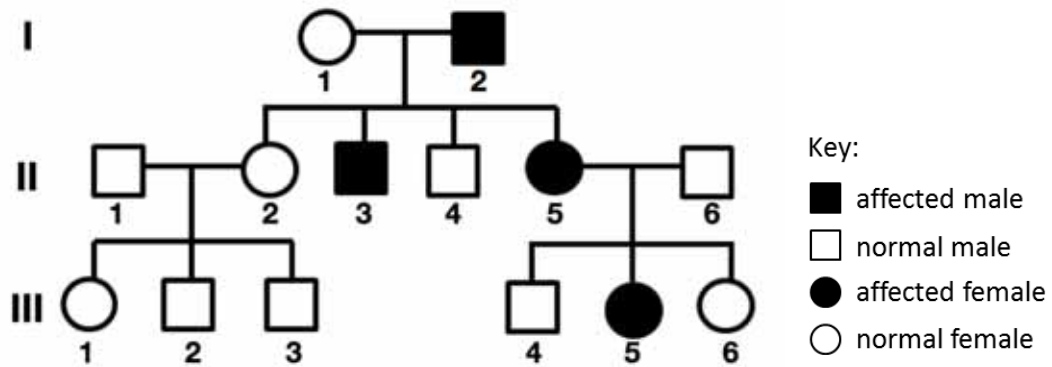


Fig. 8.1

(b) Explain the mode of inheritance of this disease that best explains the pedigree shown above. [3]

a. Autosomal dominant disease;;

Evidence for dominant

- b. Affected offspring has to inherit the dominant allele from one of his/her parents, thus one of his/her parent has to be affected too**
- c. as seen in individuals I2 and I5 (quote)**

Evidence for autosomal

- d. If the disease was sex-linked and the dominant allele was carried on the X chromosome in I2, all of his daughters would be affected.**
- e. This is because they would have inherited one of the X chromosome from him. However, this is not seen in II2. (quote)**

Question 9 [12 marks]

The sensitivity of bacteria to antibiotics can be tested using the disc diffusion method. A small number of bacteria is spread onto agar culture plates and then filter discs impregnated with antibiotics are pressed onto the surface of the agar. The plates are incubated. Bacteria grow as a 'lawn' across the agar, but a circular zone (the zone of inhibition) appears around any disc where bacterial growth is inhibited.

Two species of bacteria, **A** and **B**, were grown on separate culture plates in the presence of three types of filter paper disc:

- 1 – no antibiotics (control)
- 2 – penicillin V, a natural penicillin
- 3 – carboxypenicillin, a synthetic penicillin.

The appearance of the incubated plates is shown in **Fig. 9.1**.

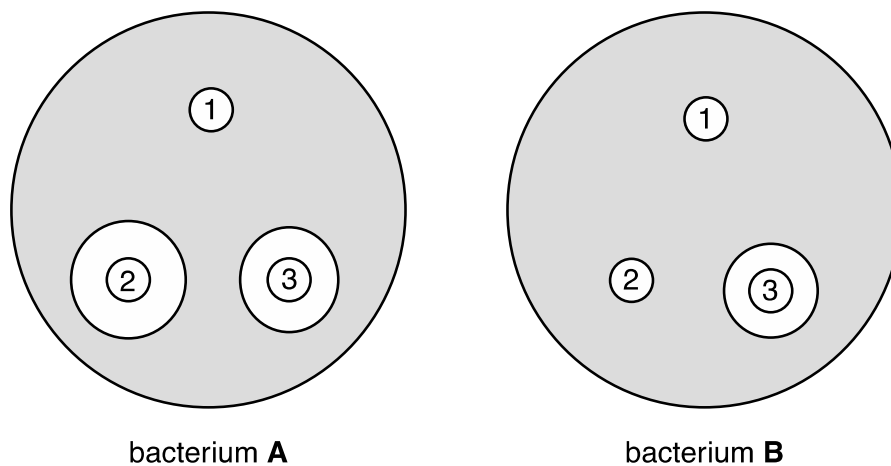


Fig. 9.1

(a) With reference to **Fig. 9.1**, explain the effects of penicillin V on bacterium **A**. [3]

- a. penicillin inhibits transpeptidase enzyme;
- b. thus preventing individual peptidoglycan chains from cross-linking / prevents formation of cross-linking between peptidoglycan chains;
- c. in growing or developing bacteria → thus bacteriostatic;
- d. penicillin also causes pore formation on the cell walls by stimulating bacterial holin proteins;

Effect:

- e. resulting in a weaker cell wall / cell unable to withstand osmotic / mechanical stress/ resulting in the cell bursting;

Compulsory point:

- f. producing a zone of zero bacterial growth around filter paper disc 2;

Bacteria **A** and **B** have different outer layers, as shown in **Fig. 9.2**.

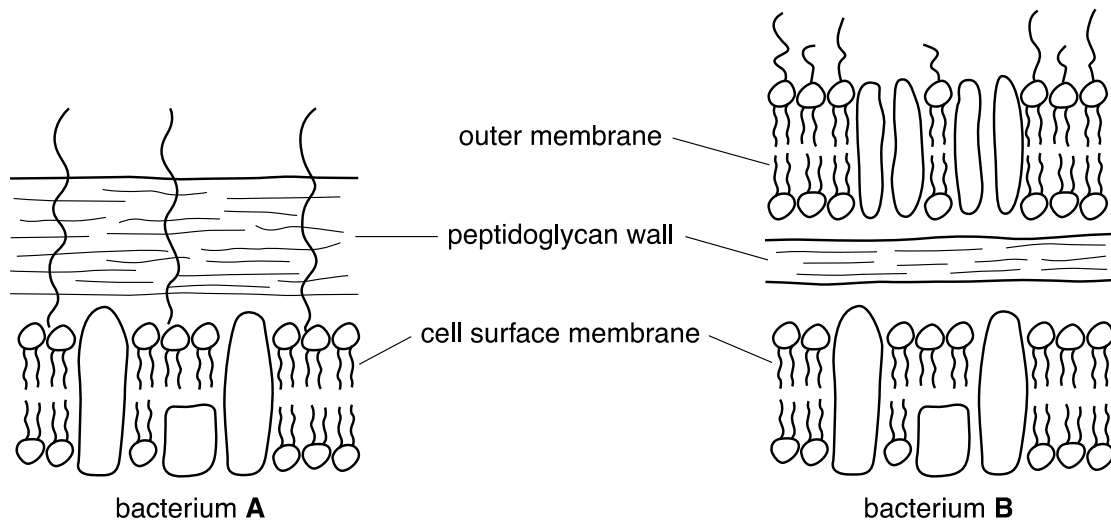


Fig. 9.2

(b) With reference to **Fig. 9.1** and **9.2**,

(i) describe how the outer layers of bacterium **B** differ from those of bacterium **A**. [2]

- a. presence of an additional outer membrane of phospholipid bilayer containing channel proteins ;;
- b. thinner peptidoglycan cell wall layer / ORA ;;

(ii) account the different effects of penicillin V on bacteria **A** and **B**. [3]

- a. zone of clearance of zero of bacteria A in filter paper 2 is larger than in B;;
- b. A more susceptible to penicillin as compared to B / ORA;
- c. as peptidoglycan cell wall in A is exposed → (idea of) penicillin can reach the wall of the bacteria;
- d. outer membrane in B prevents exposure of peptidoglycan cell wall to penicillin;
- e. and penicillin cannot diffuse through the transport proteins in outer membrane of B;

(iii) suggest how the synthetic penicillin, carboxypenicillin, is able to affect the growth of bacterium **B**. [2]

- a. drug is able to penetrate the outer membrane ;;
- b. as the drug can be transported through the channel protein due to its hydrophilic nature ;;

OR

- c. as the drug can directly pass through the phospholipid bilayer due to its hydrophobic nature ;;

- (c) Outline two other ways in which antibiotics are effective against diseases caused by bacteria. [2]

Choose any two pairs of answers:

- a. by inhibiting protein synthesis (e.g. tetracyclins);
- b. where the antibiotic inhibits the ribosome function, thus proteins essential for bacterial growth / repair / function / metabolism / survival cannot be produced;
- c. by inhibiting synthesis of nucleic acids (e.g. quinolones);
- d. which inhibits DNA gyrase / topoisomerases → causes supercoiling → no DNA replication / repair / transcription;
- e. are metabolic antagonists (e.g. sulfonamides);
- f. which inhibits folate formation → no purine formation → no dNTP for DNA synthesis;